

Hexaphotone

The Six-Dimensional History of Light

Contents

1. Abstract
2. Introduction
3. Formal definitions
4. Operator formalism for interactions
5. Mapping Hexaphoton $\rightarrow$ modal amplitudes
6. Connection to scattering theory (Mie and T-matrix) and effective transfer functions
7. Numerical implementation and reproducibility
8. Experimental predictions and test protocols
9. Discussion and extensions
10. Conclusion
11. References
12. License
13. Ethical use statement
14. Appendix

1 Abstract

This document presents a formal, self-contained formulation of the **Hexaphotone** concept: an information-centric model that represents the informational state of an electromagnetic radiation element by a six-component vector. Light fields are modeled as superpositions of information-carrying modes; interactions with matter are represented as linear operators acting on modal coefficient spaces. A general mapping from the Hexaphoton vector to modal amplitudes is defined and connected to classical scattering theory (Mie and T-matrix) to provide physically grounded reference operators. Numerical implementation steps, validation metrics and experimentally testable protocols are specified to ensure reproducibility and empirical evaluation.

2 Introduction

The Hexaphotone framework reframes standard electromagnetic field descriptors (energy, frequency, polarization, phase, momentum, direction) as a compact information vector and emphasizes modal structure and operator action as the mechanisms by which source information becomes observable images and patterns. The framework is intentionally modular: (a) a source representation (Hexaphoton ensemble), (b) a modal mapping that produces incident modal amplitudes, and (c) a material/operator model that transforms incident modes into outgoing modes. This separation enables direct comparison with classical scattering solutions and numerical solvers.

3 Formal definitions

Hexaphoton (information vector)

$$\mathbf{H} = (E, f, \mathbf{P}_i, \phi, \vec{p}, \vec{d})$$

Interpretation of components

- E — energy scale (proportional to modal energy sum).
- f — central frequency or spectral centroid.
- $\mathbf{P}_i$ — polarization state (Jones or Stokes representation).
- ϕ — global or relative phase.
- $\vec{p}$ — momentum vector.
- $\vec{d}$ — propagation direction (unit vector).

Field representation as modal superposition

$$\mathbf{E}(\mathbf{r}, t) = \sum_{k=0}^{\infty} a_k(t) * u_k(\mathbf{r})$$

Here $u_k(\mathbf{r})$ denotes an orthonormal set of spatial (and vector) modes; $a_k(t)$ are complex modal amplitudes encoding amplitude, phase and polarization.

Spatial Fourier representation

$$\tilde{\mathbf{E}}(\mathbf{k}, t) = \mathcal{F} \mathbf{E}(\mathbf{r}, t)(\mathbf{k})$$

4 Operator formalism for interactions

Material operator (general linear model)

$$\mathbf{E}_{out}(\mathbf{r}, t) = T_M[\mathbf{E}_{in}(\mathbf{r}, t)]$$

In spatial frequency representation:

$$\tilde{\mathbf{E}}_{out}(\mathbf{k}, t) = H_M(\mathbf{k}; f) * \tilde{\mathbf{E}}_{in}(\mathbf{k}, t)$$

$H_M(k; f)$ is a generally complex, matrix-valued transfer function that encodes geometry, material parameters and frequency dependence.

Measurement as projection

$$I_j = \langle m_j, \mathbf{E}_{out} \rangle = \int m_j^*(\mathbf{r}) \cdot \mathbf{E}_{out}(\mathbf{r}) d^3 r$$

Measurement outcomes are inner products of the outgoing field with detector modes m_j . Apparent state reduction is interpreted as selection of modal channels by the measurement operator.

Operator decomposition

$$T_M = T_{in} \circ T_{surf} \circ T_{bulk}$$

Components represent propagation, surface interactions (reflection/refraction) and bulk scattering. Each component admits a representation in the chosen modal basis.

5 Mapping Hexaphoton → modal amplitudes

A mapping F assigns modal amplitudes to a Hexaphoton vector:

$$a_k(\mathbf{H}) = F_k(E, f, Pi, phi, \vec{p}, \vec{d})$$

Construction principles

- **Energy normalization**

$$\sum_k \text{norm}(a_k(\mathbf{H}))^2 = C * E$$

C is a units constant.

- **Spectral weighting:** f defines a spectral envelope $S(f)$ applied to modes with frequency support near f .
- **Polarization mapping:** for vector mode components $e_{k, \alpha}$

$$a_{k, \alpha}(\mathbf{H}) = \tilde{a}_k(\mathbf{H}) * (Pi \cdot \hat{e}_{k, \alpha})$$

Phase control: phi imposes global or relative phase factors $\exp(i * phi)$.

- **Momentum / direction steering:** $\vec{p}, \vec{d}$ shift the modal distribution in k -space. A continuous prototype:

$$a(\mathbf{k}; \mathbf{H}) = A(E, f) * P(Pi) * \exp(-(\text{norm}(\mathbf{k} - \mathbf{k}_0)^2) / (2 * \text{sigma}(f)^2)) * \exp(i * phi)$$

k_0 is determined by $\vec{d}$ and $\vec{p}$, $\text{sigma}(f)$ is a frequency-dependent spread, $P(Pi)$ encodes polarization weighting.

Projection onto discrete modal basis

$$a_k(\mathbf{H}) = \int u_k^i(\mathbf{k}) * a(\mathbf{k}; \mathbf{H}) d^2 k$$

The mapping F_k is a modeling choice; different parameterizations define model variants and are subject to empirical calibration.

6 Connection to scattering theory (Mie and T-matrix) and effective transfer functions

Mie theory (sphere reference)

For a homogeneous dielectric sphere (radius R , relative refractive index m), Mie theory yields multipole coefficients $a_n(f), b_n(f)$ and an angular scattering amplitude $S(\theta, \phi; f)$.

Effective transfer function from Mie

Under paraxial or small-angle mapping, angular scattering maps to transverse wavevectors k_{perp} . A symbolic representation:

$$H_{\text{mie}}(\mathbf{k}; f) \approx \sum_{n=0}^N \alpha_n(f) * Y_n(\hat{\mathbf{k}})$$

Coefficients $\alpha_n(f)$ are constructed from a_n, b_n ; Y_n denotes angular basis functions. Numerically, compute $S(\theta, \phi)$ and sample to obtain discrete $H_M(k)$.

T-matrix generalization

The T-matrix T maps incident multipole coefficients to scattered multipole coefficients for arbitrary scatterers. Conversion between spherical harmonic coefficients and plane-wave/modal coefficients yields a matrix representation of T_M in the modal basis.

Operator consistency

Both Hexaphoton operator model and Mie/T-matrix operators are linear maps on modal coefficient spaces. Composition:

$$a_{\text{out}}(\mathbf{k}) = H_M(\mathbf{k}) * a_i(\mathbf{k})$$

with a_i produced by the Hexaphoton mapping and H_M provided by Mie/T-matrix or numerical solver.

7 Numerical implementation and reproducibility

Computational pipeline (concise)

1. Define Hexaphoton ensemble H_i and construct continuous modal distributions $a(k; H_i)$.
2. Discretize k -space and choose modal basis u_k .
3. Compute incident field via inverse Fourier transform of $a(k)$.
4. Compute object operator: for a sphere use Mie code to obtain $S(\theta, \phi)$; map to $H_M(k)$. For general objects use T-matrix or numerical solvers (BEM, FEM, FDTD).
5. Apply operator: $a_{out}(k) = H_M(k) * a_{in}(k)$.
6. Reconstruct outgoing field and compute observables.

Suggested validation metrics

- **Spatial mean squared error (MSE)**

$$MSE = (1/N) * \sum_{i=1}^N (I_{ref}(\mathbf{r}_i) - I_{model}(\mathbf{r}_i))^2$$

Spectral overlap: normalized inner product of Fourier spectra.

- **Polarization fidelity:** angular difference and degree of polarization comparison across the field.

Software recommendations

- Numerical libraries: *NumPy*, *SciPy*, *Matplotlib*.
- Mie computations: established Mie libraries (e.g., *miepython* or equivalent).
- T-matrix: available T-matrix packages for spheroids and aggregates.
- Reproducibility: provide notebooks, parameter files and *requirements.txt* with exact versions.

Discretization and sampling notes

- Ensure k -space sampling satisfies Nyquist criteria for the spatial domain of interest.
- Map angular Mie outputs to transverse k via $k_{perp} = (2 * \pi / \lambda) * \sin(\theta)$ in paraxial approximation.

8 Experimental predictions and test protocols

Prediction A — Reflection imaging of shaped surfaces

A shaped reflective surface illuminated by a Hexaphoton ensemble with directional and modal structure will produce a projected intensity distribution that preserves contour information when T_{refl} transmits the relevant modal channels. Measurement: compare projected intensity with $T_{\text{refl}}[a(k)]$ predictions.

Prediction B — Slit filtering and source dominance

A narrow aperture suppresses high spatial frequencies; transmitted patterns emphasize low-frequency components that reflect source structure. Measurement: record transmitted intensity for different aperture geometries and compare Fourier spectra to incident modal distribution.

Prediction C — Sphere reference test

A homogeneous dielectric sphere acts as a known operator H_{mie} . For incident Hexaphoton distributions with controlled $\vec{d}$ and $\vec{p}$, the scattered angular distribution should match Mie-based predictions when modal mapping F is applied. Measurement: angularly resolved scattering and near-field intensity mapping.

Protocol essentials

- Use narrowband and broadband sources to separate spectral effects.
- Record polarization-resolved intensity.
- Calibrate detectors and correct for system transfer functions.
- Provide raw data and processing scripts for reproducibility.
- Account for detector and system transfer functions when comparing model to measurements.

9 Discussion and extensions

The Hexaphotone framework is an information-centric modeling layer that interfaces source characterization and classical scattering/propagation operators. The mapping F is the principal modeling choice; its parameterization determines predictive power and must be constrained by experiment or simulation. The connection to Mie and T-matrix theory provides physically grounded benchmark operators.

Limitations

- The present formulation is classical; single-photon quantum effects require field quantization and operator expectation values.
- Nonlinear and inelastic interactions are outside the linear operator formalism.
- Modal basis choice affects numerical efficiency and interpretability.

Extensions

- Include orbital angular momentum and topological mode indices as additional information channels.
- Extend F to stochastic ensembles to model partial coherence.
- Promote modal amplitudes to annihilation/creation operators for quantum extensions and define Hexaphoton states in Fock space.

10 Conclusion

Hexaphotone Theory formalizes an information-centric representation of light and provides a modular operator framework to relate source information to observable fields via established scattering theory. The mapping and numerical pipeline enable reproducible comparisons with Mie/T-matrix benchmarks and yield experimentally testable predictions. The framework is modular and extensible for further theoretical and experimental development.

11 References

- J. W. Goodman, *Introduction to Fourier Optics*, 3rd ed.
- M. Born and E. Wolf, *Principles of Optics*, 7th ed.
- C. F. Bohren and D. R. Huffman, *Absorption and Scattering of Light by Small Particles*.
- Standard references on T-matrix methods and vector spherical harmonics.

12. License

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14. Appendix: central equations in LibreOffice-Math syntax (copyable)

Hexaphoton vector

$$\mathbf{H} = (E , f , \Pi , \phi , \mathbf{p} , \mathbf{d})$$

Modal expansion

$$\mathbf{E} (\mathbf{r} , t) = \sum_{k=0}^{\infty} \{ a_k (t) * u_k (\mathbf{r}) \}$$

Fourier transform

$$\tilde{\mathbf{E}} (\mathbf{k} , t) = F \{ \mathbf{E} (\mathbf{r} , t) \} (\mathbf{k})$$

Material transfer in k-space

$$\tilde{\mathbf{E}}_{\text{out}} (\mathbf{k} , t) = H_M (\mathbf{k} ; f) * \tilde{\mathbf{E}}_{\text{in}} (\mathbf{k} , t)$$

Measurement projection

$$I_j = \langle m_j , \mathbf{E}_{\text{out}} \rangle = \int \{ m_j^* (\mathbf{r}) \cdot \mathbf{E}_{\text{out}} (\mathbf{r}) d^3 r \}$$

Gaussian prototype for a(k;H)

$$a (\mathbf{k} ; \mathbf{H}) = A(E , f) * P(\Pi) * \exp(- (\text{norm}(\mathbf{k} - \mathbf{k}_0)^2) / (2 * \text{sigma}(f)^2)) * \exp(i * \phi)$$

Projection onto discrete modes

$$a_k (\mathbf{H}) = \int \{ u_k^* (\mathbf{k}) * a (\mathbf{k} ; \mathbf{H}) d^2 k \}$$

Operator composition

$$T_M = T_{\text{prop}} \circ T_{\text{surf}} \circ T_{\text{bulk}}$$

Application of operator

$$a_{\text{out}} (\mathbf{k}) = H_M (\mathbf{k}) * a_{\text{in}} (\mathbf{k})$$

Intensity and MSE

$$I (\mathbf{r}) = \text{norm}(\mathbf{E} (\mathbf{r}))^2$$

$$\text{MSE} = (1 / N) * \sum_{i=1}^N \{ (I_{\text{ref}} (\mathbf{r}_i) - I_{\text{model}} (\mathbf{r}_i))^2 \}$$

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